

Square ValuesWorksheet 7A.02.01 **ANSWER KEY**

1. Complete the table. Each area is the side length multiplied by itself.

Side s	1	2	3	4	5	6	7	8	9	10
Area s^2	1	4	9	16	25	36	49	64	81	100

These ten values are the perfect squares students should recognize on sight. A student who writes 20 under $s = 10$ has multiplied by 2 instead of squaring — the single most common error on this benchmark.

2. Select ALL of the numbers below that are perfect squares.

- | | |
|---|--|
| ☒ 1 | ☐ 8 |
| ☒ 16 | ☐ 20 |
| ☒ 36 | ☐ 45 |
| ☒ 49 | ☐ 60 |
| ☒ 64 | ☒ 100 |

Six of the ten, 8, 20, 45 and 60 are the distractors: each is close to a perfect square but is not one. Ask for the whole number that was squared — a student who cannot name it has not shown the answer.

For questions 3 through 5, find the area of a square with the side length given.

3. Side length 12 units

$$\text{Area} = 12^2 = 12 \times 12 = 144 \text{ square units}$$

4. Side length $\frac{2}{5}$ cm

$$\text{Area} = \frac{2}{5} \times \frac{2}{5} = \frac{4}{25} \text{ cm}^2$$

Square the numerator and square the denominator. The answer is not $4/5$.

5. Side length 0.7 units

$$\text{Area} = 0.7 \times 0.7 = 0.49 \text{ square units}$$

Two decimal places in, two out. A student who answers 4.9 has multiplied by 7 rather than 0.7.

For questions 6 through 8, find the length of one side of a square with the area given.

6. Area 81 square inches

$9 \times 9 = 81$, so the side is 9 inches.

7. Area $\frac{4}{25}$ square centimeters

$2 \times 2 = 4$ and $5 \times 5 = 25$, so the side is $\frac{2}{5}$ cm.

8. Area 2.25 square units

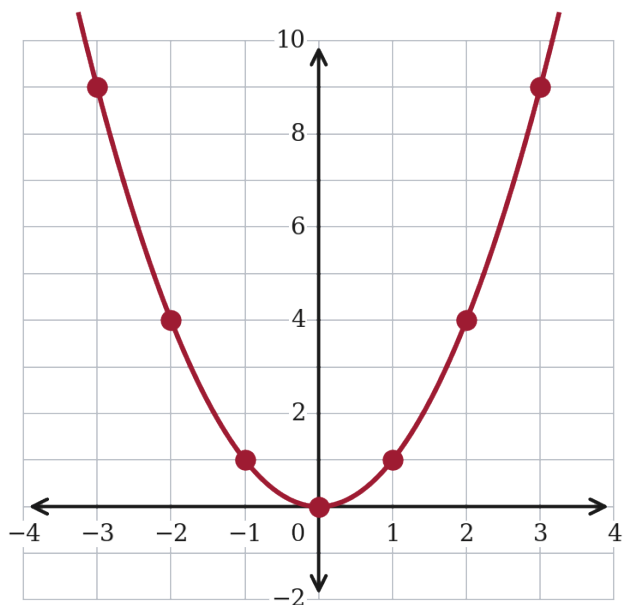
$1.5 \times 1.5 = 2.25$, so the side is 1.5 units.

A side length is a distance, so only the positive value is used here. The negative one, -1.5 , arrives in Lesson 2, where the question becomes $x^2 = 2.25$ instead of “what is the side?”

Questions 9 and 10 use the equation $y = x^2$.

9. Complete the table of values. Then plot the seven points on the grid.

x	Substitute into $y = x^2$	Coordinates (x, y)
-3	$y = (-3)^2 = 9$	$(-3, 9)$
-2	$y = (-2)^2 = 4$	$(-2, 4)$
-1	$y = (-1)^2 = 1$	$(-1, 1)$
0	$y = (0)^2 = 0$	$(0, 0)$
1	$y = (1)^2 = 1$	$(1, 1)$
2	$y = (2)^2 = 4$	$(2, 4)$
3	$y = (3)^2 = 9$	$(3, 9)$



Opposite x-values give the same y-value, which is the whole reason the graph is symmetric.

10. Will the graph of $y = x^2$ ever go below the x-axis? Explain why or why not.

No. A positive number times itself is positive, and a negative number times itself is also positive.

The only value of x that gives $y = 0$ is $x = 0$, so the graph touches the x-axis at $(0, 0)$ and is above it everywhere else.

The point requires both halves: why y is never negative, AND what happens at $x = 0$. A student who writes “the graph is always positive” has missed the origin.

11. Can a square have an area of 8 square units? If it can, name the two whole numbers its side length falls between. Explain how you know.

Yes. The side is between 2 and 3.

$2 \times 2 = 4$, which is too small, and $3 \times 3 = 9$, which is too big. So the side length is bigger than 2 and smaller than 3.

The side is not a whole number.

Do not accept “the side is a decimal or a fraction.” It is neither — but that is Lesson 4’s work, and students are not expected to say so yet. The point needs 2 and 3 identified and the two products given as the reason.